

A minimal counterexample to Class 69 mesh-pattern equidistribution on involutions

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Abstract

We give a minimal counterexample to the involution equidistribution assertion in Conjecture 1 of Fang, Fu, Kitaev, Li, Su, and Sun, arXiv:2606.14367v1. For its first and third mesh patterns, the occurrence polynomials on the four involutions of size three are respectively $2 + y + y^2$ and $1 + 2y + y^2$. The distributions agree in every smaller size. A complete inspection of seven increasing pairs proves these statements without a computational dependency. The correction concerns the explicitly identified v1 assertion; no claim of priority is made.

1 Patterns and statement

Let S_n be the permutations of $\{1, \dots, n\}$, written in one-line notation, and let $\mathcal{I}_n = \{\pi \in S_n : \pi^2 = \text{id}\}$; fixed points are allowed. We include the empty permutation in $\mathcal{I}_0$. Consider the length-two mesh patterns

$$\begin{aligned} P &= (12, R_P), \quad R_P = \{(0, 0), (1, 1), (1, 2), (2, 1)\}, \\ Q &= (12, R_Q), \quad R_Q = \{(0, 2), (1, 0), (1, 1), (2, 1)\}. \end{aligned} \tag{1}$$

As in Definition 2.1 of [1], a shaded unit cell is indexed by its bottom-left corner (Figure 1). An increasing pair at positions $i < j$, with $a = \pi_i < b = \pi_j$, is an occurrence of $(12, R)$ if every other point (t, π_t) has cell index

$$c_{i,j}(t) = (\mathbf{1}_{\{t>i\}} + \mathbf{1}_{\{t>j\}}, \mathbf{1}_{\{\pi_t>a\}} + \mathbf{1}_{\{\pi_t>b\}}) \notin R. \tag{2}$$

Here $\mathbf{1}$ denotes the indicator. This is exactly the empty-open-cell convention: the selected positions and values divide each axis into three intervals, and no other point lies on a dividing line. The selected points themselves are excluded from the test.

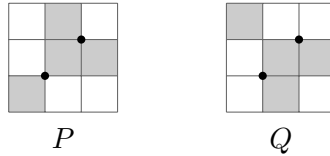


Figure 1: The patterns in (1). Each grid runs from 0 to 3 on both axes, with dots at $(1, 1)$ and $(2, 2)$; the vertical coordinate increases upward.

Write $\text{occ}_M(\pi)$ for the number of occurrences of a pattern M , and set

$$F_n^M(y) = \sum_{\pi \in \mathcal{I}_n} y^{\text{occ}_M(\pi)}.$$

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Equidistribution on involutions means $F_n^P(y) = F_n^Q(y)$ for every n , or equivalently equality of the numbers having k occurrences for every $n, k \geq 0$.

Theorem 1. *For the patterns in (1),*

$$F_3^P(y) = 2 + y + y^2, \quad F_3^Q(y) = 1 + 2y + y^2.$$

Moreover, $F_n^P = F_n^Q$ for $n = 0, 1, 2$, so size three is minimal. In particular, Conjecture 1 as displayed in arXiv:2606.14367v1 is false.

2 Complete calculation

Proof. Every involution is a product of disjoint transpositions and fixed points. On three letters there can be at most one transposition, so the complete list is

$$\mathcal{I}_3 = \{123, 132, 213, 321\}.$$

For an increasing pair there is a unique remaining point, and (2) gives its cell. All seven increasing pairs are listed below. A check mark means that the remaining cell is absent from the relevant shading set and hence the pair is an occurrence.

π	Index pair (i, j)	Remaining cell	P	Q
123	(1, 2)	(2, 2)	✓	✓
	(1, 3)	(1, 1)	—	—
	(2, 3)	(0, 0)	—	✓
132	(1, 2)	(2, 1)	—	—
	(1, 3)	(1, 2)	—	✓
213	(1, 3)	(1, 0)	✓	—
	(2, 3)	(0, 1)	✓	✓

The permutation 321 has no increasing pair. Summing the two occurrence columns for each permutation gives

π	123	132	213	321
$\text{occ}_P(\pi)$	1	0	2	0
$\text{occ}_Q(\pi)$	2	1	1	0

and therefore the asserted polynomials. In particular, the P -avoiders are 132 and 321, whereas the only Q -avoider is 321. Thus the universally quantified equality already fails at $(n, k) = (3, 0)$.

For $n = 0, 1$ there is a unique involution and no length-two occurrence, so both polynomials are 1. For $n = 2$, the involutions are 12 and 21. The first has one increasing pair and no remaining points, while the second has no increasing pair. Both polynomials are consequently $1 + y$. This proves minimality for all occurrence counts k .

Finally, the first and third diagrams in Conjecture 1 of [1] have dots at $(1, 1), (2, 2)$ and shaded lower-left corners R_P, R_Q , respectively. Its assertion that all four displayed patterns are equidistributed on involutions would imply $F_3^P = F_3^Q$. The calculation above disproves that assertion in the specified version. $\square$

3 Context, scope, and reproducibility

The nearest all-permutation result is Theorem 1.1 of Zhang and Zhao [2], which establishes the remaining Class 69 distribution-equivalence over S_n . That statement concerns sums over all permutations. It does not assert the corresponding equality after restriction to $\mathcal{I}_n$. Theorem 1 supplies an explicit obstruction to that restriction. As a finite consistency check, direct enumeration of all six permutations in S_3 gives $2 + 3y + y^2$ for each of the four diagrams in the cited conjecture. This finite check is not a proof of an all-size identity.

Exact enumeration and occurrence polynomials place this calculation in the broader setting of certified permutation enumeration; Zeng’s study of distinct cycle lengths [3] provides adjacent context for exact coefficient verification. No result from that work is used in the proof above.

The source comparison is with the saved v1 texts of [1, 2]. Priority, earlier corrections, later revisions, and current publication status have not been established. The mathematical conclusion concerns the two explicitly specified patterns and the displayed v1 involution assertion. It makes no claim about a revised conjecture, a new bijection, asymptotics, or a broader classification.

Computation and assistance. The accompanying standard-library Python program `verify.py` enumerates all permutations for $0 \leq n \leq 3$, filters by $\pi_{\pi_i} = i$, and computes occurrences using (2). It checks the complete $\mathcal{I}_3$ table, the smaller-size polynomials, and the four-diagram S_3 control using only integer arithmetic. The displayed proof is self-contained and does not rely on execution of the program. OpenAI Codex assisted with the finite search, source-diagram transcription, verification code, and manuscript preparation. A separate model-context review recomputed the finite certificate and inspected the source diagrams; this was not external peer review or formal proof checking.

References

- [1] Qi Fang, Shishuo Fu, Sergey Kitaev, Haijun Li, Xinyu Su, and Ziyao Sun, *On mesh patterns of short length: Equidistribution and enumeration*, arXiv:2606.14367v1 (2026). <https://arxiv.org/abs/2606.14367v1>.
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- [3] Zijian Zeng, *The Distinct-Cycle-Length Probability is Strictly Decreasing after 30*, 2026. <https://doi.org/10.2139/ssrn.7380519>.